PROJECT DEFENSE & TECHNICAL Q&A MASTER GUIDE

RecoverIQ: AI Loan Collections Prioritiser · Candidate: Nahbila Faith · Track: SKY8-DS-FP-005

LIVE PRODUCTION APP:
https://zshops-indicators-despite-muscle.trycloudflare.com
Port 8080 · 15/15 Automated Unit Tests Passed

1. Technical Architecture & Algorithms Used in RecoverIQ

RecoverIQ is not a simple heuristic script or an ungrounded LLM. It is an end-to-end multi-task machine learning system designed specifically for the conditional distribution **P(Recovery | Arrears >= 15 Days)**. Below are the 5 algorithms powering the engine:

Component / Task	Exact Algorithm & Architecture	Mathematical Formula / Objective	Role in the Product
1. Primary Recovery Classifier	**HistGradientBoostingClassifier** with **CalibratedClassifierCV** (Sigmoid/Platt Scaling, 3-fold CV). 180 trees, lr=0.045, max_leaf_nodes=31, balanced weights.	$\min \Sigma -[y \log p + (1-y)\log(1-p)]$ $p = \sigma(a \cdot f(x) + b)$	Outputs calibrated probability **P(cure=1)** that an account in arrears will pay >=95% within 30 days.
2. Time-to-Cure Estimator	**HistGradientBoostingRegressor** (Survival Hazard Regressor). 120 trees, lr=0.045, max_leaf_nodes=25.	y_hat = clamp(round(f(x)), 1, 30)	Predicts exact expected business days until settlement (used to calculate daily velocity).
3. Partial Settlement Ratio	**HistGradientBoostingRegressor** with L2 regularization=1.0. 100 trees, lr=0.05.	r_hat = clamp(f(x), 0.35, 1.0)	Predicts expected ratio of overdue principal recovered (accounts for partial payments).
4. Causal Uplift & Persuadability	**Two-Model Treatment Effect Engine** (Individual Treatment Effect / Uplift Modeling).	ITE = P(Y=1 | T=1) - P(Y=1 | T=0)	Segments borrowers into: Persuadables, Self-Curers, Sleeping Dogs, and Lost Causes.
5. Explainable AI (XAI) Waterfall	**Additive Tree SHAP / Marginal Feature Contribution Engine**.	Contribution_i = Δ log-odds relative to population baseline	Generates the positive and review signal badges shown on the officer's Member Action Desk.

2. Technical & Modeling Defense Q&A (Expect from ML Judges)

Q1: Why did you choose Gradient Boosting instead of Logistic Regression or a Deep Neural Network?

Direct Answer: Tabular financial data with skewed distributions, non-linear payment thresholds, and missing values is where Gradient Boosted Decision Trees (GBDT) consistently outperform Deep Neural Networks. GBDT handles correlated features (like past days late and total late count) without distortion, doesn't require fragile feature normalization, and natively handles missing values. Compared to simple Logistic Regression, it captures non-linear interactions (e.g. high debt interacting with short arrears duration vs high debt with long arrears duration).

Defense Pro-Tip: Mention that HistGradientBoosting is optimized with histogram binning (O(n) complexity), allowing lightning-fast inference under 2 milliseconds on a lightweight CPU server.

Q2: How did you prevent data leakage? Can you prove post-contact data didn't enter the model?

Direct Answer: Zero future leakage was strictly designed into the temporal pipeline and enforced by automated testing. All features are strictly anchored to the observation timestamp T0 (Days_Instalment + 15). Features only use historical records where Days_Instalment < T0. Any post-contact field (like contact outcome, promises kept, or post-contact payment timestamps) is completely excluded from feature engineering. We wrote automated unit test `tests/test_no_leakage.py` that asserts no target or post-contact field enters the feature column matrix, passing 100%.

Defense Pro-Tip: State clearly: 'An officer cannot know what happens during a call before dialing. Our test suite guarantees the model only sees pre-call history.'

Q3: Why is your target formulated as 'Recovery Given Arrears' instead of Standard Default?

Direct Answer: Standard credit risk models predict P(Default) across all applicants at loan origination. In loan collections, default has already occurred—the account is already 15+ days late. Training on all borrowers would answer the wrong question and bias the model toward underwriting criteria (income, age). Our model isolates the conditional distribution P(Recovery | Overdue >= 15 Days). Out of 2.5 million records, we isolated exactly the 36,934 delinquent events, where the unassisted base recovery rate is only 11.94%.

Defense Pro-Tip: Refer to Section 2 of the Project Brief: 'Getting that distinction right is most of this project.'

Q4: Why did you use Precision@50 and Precision@100 instead of ROC-AUC or Overall Accuracy?

Direct Answer: In our operational setting, a branch has 400 arrears accounts but only 3 collections officers. Three officers cannot work 400 accounts in a single week; they can feasibly handle 50 to 100 deep contacts. Overall accuracy is misleading because an empty model predicting 'no one will pay' would achieve 88% accuracy due to class imbalance (11.9% base cure). ROC-AUC evaluates the entire curve, but we only care about the top of the queue where officers actually spend time. At top-100 accounts, our model achieves 88.0% precision vs 7.0% for biggest-debt-first—a 12.6x lift.

Defense Pro-Tip: Emphasize: 'Officers don't work the whole curve; they work the top 100 accounts. Precision@100 directly equals cash in the vault.'

Q5: How do you handle class imbalance if only 11.9% of accounts cure?

Direct Answer: We used three combined techniques: First, `class_weight='balanced'` in HistGradientBoosting, which penalizes false negatives inversely proportional to class frequencies. Second, Platt Scaling calibration (`CalibratedClassifierCV` with 3-fold cross-validation) so that the raw logit scores represent true, well-calibrated probabilities. Third, we rank the call roster not by binary 0/1 labels, but by Expected Value: EV = P(recovery) × Partial_Ratio × Debt.

Defense Pro-Tip: Show that Platt scaling ensures that an account with 0.70 predicted probability actually cures 70% of the time in real life.

3. Business Model, Commercialization & Pitch Q&A (Judges & Buyers)

Q6: Why did you price the software at 35,000 FCFA/month instead of a huge enterprise price?

Direct Answer: In the Cameroonian microfinance sector (NtaCCUL, AZICCUL, community unions), any expense above 100,000 FCFA requires board authorization and central committee meetings that can take 6 months. At 35,000 FCFA/month (~1,160 FCFA/day), a local Branch Manager has discretionary petty-cash authority to sign off immediately. Furthermore, 35,000 FCFA is less than what one officer spends on motorcycle fuel. Recovering just one single small overdue loan of 35,000 FCFA pays for the entire monthly subscription, making adoption frictionless.

Defense Pro-Tip: Highlight: 'Low friction enables us to onboard 10 branches in 60 days rather than chasing 1 contract for a year.'

Q7: How much does it cost you as the developer to run this platform?

Direct Answer: Because our machine learning models run fast local inference in Python without paid third-party AI APIs (no per-token OpenAI charges), our monthly hosting cost is extremely small: Hetzner Cloud Linux VPS (2 vCPU, 4GB RAM) is 5,400 FCFA ($9), domain name is 600 FCFA ($1), and developer 4G monitoring bundle is 10,000 FCFA ($16.60). Total monthly out-of-pocket expenditure is only ~16,000 FCFA ($26.60). The credit union pays for their own bulk SMS airtime directly through their corporate telecom accounts.

Defense Pro-Tip: Show the unit economics: 'One single branch paying 35,000 FCFA covers 218% of my total server costs, giving instant cash-flow profitability.'

Q8: If you're only charging 35,000 FCFA/month, how do you make serious money?

Direct Answer: Through multi-branch network scaling and upfront onboarding fees. While 35,000 FCFA is small for one branch, the CamCCUL network and NtaCCUL operate dozens of branches across Mezam Division. When we onboard 10 branches, recurring revenue is 350,000 FCFA/month against 20,000 FCFA in server costs, yielding 330,000 FCFA in clean monthly profit (4 million FCFA/year). In addition, we charge a one-time setup and staff training fee of 60,000 FCFA per branch, providing immediate upfront cash flow upon onboarding.

Defense Pro-Tip: Mention that the same single $9 server can handle over 30 credit union branches simultaneously without extra hosting cost.

Q9: What is the COBAC connection, and why does a Credit Union Manager care about it?

Direct Answer: Under Central African Banking Commission Regulation COBAC R-2017/07, when an overdue loan reaches 31 days late (Category 2 arrears), the microfinance institution is legally mandated to freeze 20% of the loan amount into non-interest-earning statutory reserves. At 91 days, it is 50%, and at 180 days, 100%. Across our 400 sample accounts, resolving accounts at Day 15 during the grace window prevents 30,000,000 FCFA in capital from being locked away in reserves, keeping that money active in the vault to issue new interest-earning loans.

Defense Pro-Tip: This is your biggest selling point for Branch Managers: you are not just collecting debt; you are protecting their statutory solvency ratios.

Q10: Why can't a Credit Union just do this in Microsoft Excel?

Direct Answer: Excel can easily sort by debt size or days late, but it cannot compute multi-variate conditional probabilities. A borrower might have a large debt but also have a perfect record of keeping past promises, while another has a small debt but has missed 4 consecutive promises. Excel cannot weight actions by cost (e.g. deciding whether a 5,000 FCFA field visit makes financial sense vs a 50 FCFA SMS nudge), nor can it perform geographic clustering to route motorbike visits across Bamenda neighborhoods to save fuel.

Defense Pro-Tip: Conclude with: 'Excel is a static ledger; RecoverIQ is a dynamic decision co-pilot.'

Q11: How does the system prevent harassment of vulnerable members?

Direct Answer: Harassment in microfinance usually occurs when officers have no data, panic at end-of-month targets, and aggressively pursue the largest debtors. RecoverIQ embeds automated hardship and elder safeguards: senior members (age 60+) and members experiencing documented distress (illness, bereavement) are flagged with green safeguard badges. The system automatically blocks aggressive collection scripts, directs officers to a supportive tone, and activates the 'Small-small' daily repayment slider and grace period options.

Defense Pro-Tip: This addresses Stage E of the project brief: 'Be explicit about the human cost.'

Q12: What happens if borrower data is missing or incomplete in local credit union records?

Direct Answer: RecoverIQ implements an automated domain-informed imputation pipeline (`src/features.py`). If a borrower has no prior delinquency history, the system doesn't crash; it imputes baseline ratios, applies conservative uncertainty bounds to recovery probabilities, and surfaces a 'Review' signal badge on the officer's screen prompting them to confirm details before taking action.

Defense Pro-Tip: Our test suite tests this edge case in `test_missing_data_defense`, which passes cleanly.

Quick Numerical Cheat-Sheet for Q&A Defense:

Metric / Indicator	Incumbent (Old Way)	RecoverIQ (Our System)	Direct Benefit to Branch
Precision at Top 50 Accounts	14.0% (7 loans cured)	**90.0% (45 loans cured)**	**6.4x more recovered accounts**
Precision at Top 100 Accounts	7.0% (7 loans cured)	**88.0% (88 loans cured)**	**12.6x more recovered accounts**
Cash Collected (Top 100)	3,301,388 FCFA	**5,465,544 FCFA**	**+2,164,156 FCFA extra cash (+65.6%)**
Monthly Motorbike Fuel Spend	420,000 FCFA (Criss-crossing)	**218,000 FCFA (Route clustered)**	**-202,000 FCFA saved on petrol (-48%)**
COBAC Statutory Capital Frozen	30,000,000 FCFA locked away	**0 FCFA locked (Resolved at Day 15)**	**30,000,000 FCFA kept in vault**
Software Subscription Cost	N/A	**35,000 FCFA / month**	**Pays for itself in under 4 hours**
Developer Monthly Server Cost	N/A	**~16,000 FCFA / month ($26.60)**	**Profitable with just 1 client branch**